



DPP SOLUTION

Physical

- **Chemistry**
- **Some Basic Concepts of Chemistry**

DPP No.- 01



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Question



Chlorine atom differs from chloride ions in the number of



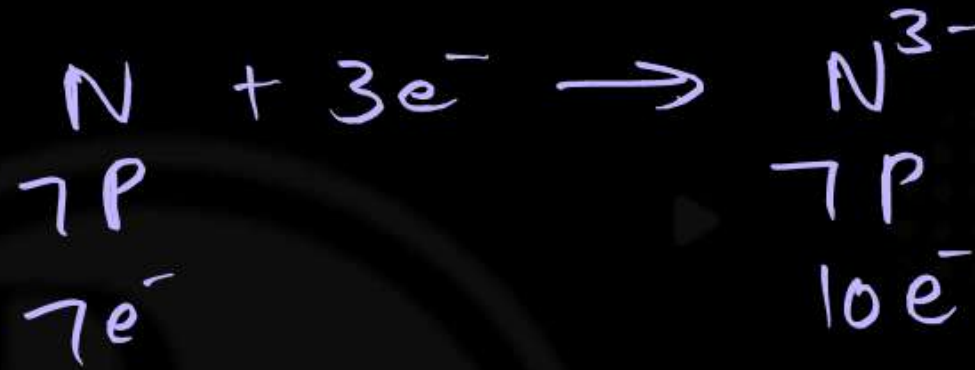
- 1** Proton
- 2** Neutron
- 3** Electrons
- 4** Protons and electrons

Question



The nitrogen atom has 7 protons and 7 electrons, the nitride ion (N^{3-}) will have

- ☒ 1 7 protons and 10 electrons
- ☐ 2 4 protons and 7 electrons
- ☐ 3 4 protons and 10 electrons
- ☐ 4 10 protons and 7 electron



Question



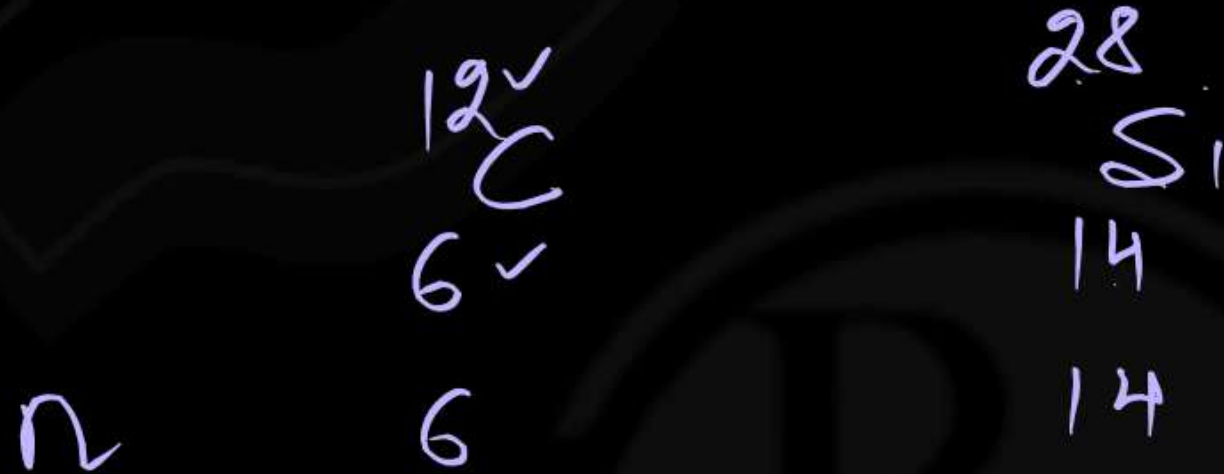
The ratio between the neutrons in C and Si with respect to atomic masses 12 and 28 is

1 $2 : 3$

2 $3 : 2$

3 $3 : 7$

4 $7 : 3$



$$\frac{6}{14} = \frac{3}{7}$$

Question

Which of the following atom has more electrons than neutrons

- 1 C
- 2 F^-
- 3 O^{2-}
- 4 Al^{3+}

e^-	n
6	6
10	10
10	8
10	14

12
C
6

19
F
9

16
O
8

27
Al
13



Question



The unified atomic mass unit has a value of

- 1 $1.661 \times 10^{-27} \text{ g}$
- 2 $1.661 \times 10^{-27} \text{ kg}$
- 3 $1.661 \times 10^{-25} \text{ g}$
- 4 $1.661 \times 10^{-25} \text{ kg}$

Question



The modern atomic weight scale is based on

- ☒ 1 C^{12}
- ☐ 2 O^{16}
- ☐ 3 H^1
- ☐ 4 C^{13}

Question



1 amu is equal to

- ☒ **1** $\frac{1}{12}$ of C – 12
- ☐ **2** $\frac{1}{14}$ of O – 16
- ☐ **3** 1 g of H₂
- ☐ **4** 1.66×10^{-23} kg

Question



Which is not a basic postulate of Daltons atomic theory?

- 1 Atoms are neither created nor destroyed in a chemical reaction
- 2 Atoms of different elements have different masses
- 3 Atoms of an element may be different due to presence of isotopes
- 4 Each element is composed of extremely small particles called atoms

Question



Choose the Incorrect Statement about Dalton's Atomic Theory

- 1** All the atoms of a given element have identical properties including identical mass.
- 2** Chemical reactions involve reorganization of atoms
- 3** ✓ Compounds are formed when atoms of different elements combine in any ratio.
- 4** Matter consists of indivisible atoms.



Thank

You...

